
Do Explanations Survive Attacks?

Adversarial Robustness of Saliency-Based Interpretability

A Mini Research Problem bridging

Lecture 07 (Robustness) x Lecture 09 (Interpretability)

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model: torchvision resnet50 (IMAGENET1K_V2) | 12 ImageNet images | code + demo linked on last slide

- Motivation: deep classifiers are deployed with saliency-map explanations (Grad-CAM, Integrated Gradients) to justify decisions, e.g. in medicine and autonomous driving. But are those explanations trustworthy when an adversary is present?
- Main idea: take a pretrained ImageNet classifier; craft imperceptible adversarial perturbations (FGSM / PGD); then measure (a) does the prediction flip, (b) does the explanation survive, and (c) can the explanation's own fragility be used to DETECT the attack?
- Results on 12 images (ResNet-50):
 - PGD flips up to 100% of predictions with an L-inf budget invisible to the eye.
 - Explanations are fragile: only ~22% of the top-salient region survives the attack (Grad-CAM IoU 0.22, IG 0.23).
 - That fragility is a signal: a noise-instability detector separates clean vs. adversarial inputs with ROC-AUC = 0.91.

Background: two ideas from the course

- Adversarial example (Lecture 07 - Robustness): an input $x' = x + \delta$ with a tiny L-inf-bounded perturbation δ that a human cannot see, yet the model mis-classifies with high confidence.
- Saliency explanation (Lecture 09 - Interpretability): a heatmap over the input showing which pixels drive the prediction - Grad-CAM (gradients x activations of the last conv layer) and Integrated Gradients (path integral of gradients).
- Research question: explanations are sold as a transparency / debugging tool. If an imperceptible attack can silently relocate the explanation, the tool is unreliable exactly when we need it. Can we quantify this - and turn it into a defense?

Related work (1/2): attacks & explanations

- Adversarial attacks
 - Szegedy et al. 2014, 'Intriguing properties of neural networks' - discovered adversarial examples. arXiv:1312.6199
 - Goodfellow et al. 2015, 'Explaining and Harnessing Adversarial Examples' - FGSM, the linear view. arXiv:1412.6572
 - Madry et al. 2018, 'Towards Deep Learning Models Resistant to Adversarial Attacks' - PGD + adversarial training. arXiv:1706.06083
 - Carlini & Wagner 2017, strong C&W attacks. arXiv:1608.04644
- Attribution / explanation methods
 - Simonyan et al. 2014, gradient saliency maps. arXiv:1312.6034
 - Selvaraju et al. 2017, 'Grad-CAM'. arXiv:1610.02391
 - Sundararajan et al. 2017, 'Integrated Gradients' (Axiomatic Attribution). arXiv:1703.01365

Related work (2/2): fragility & detection

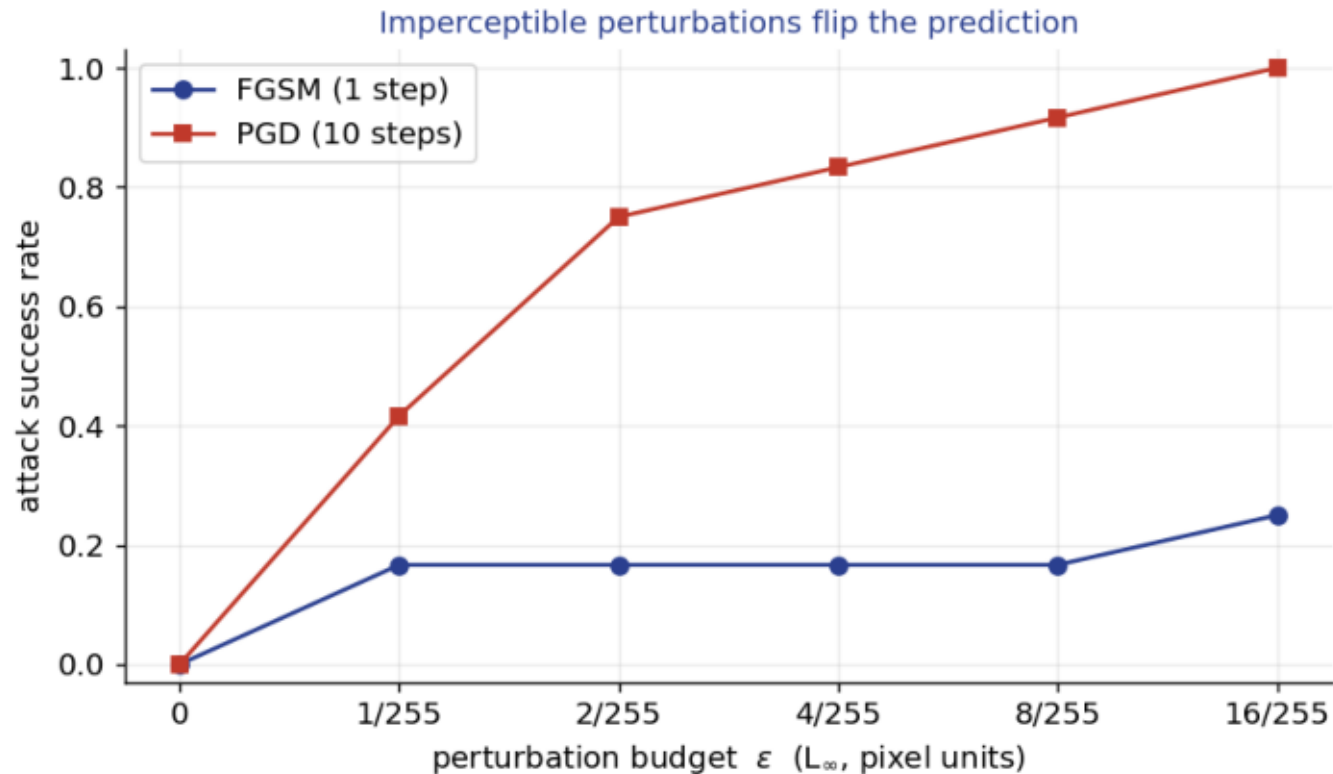
- Fragility of explanations (this MRP builds on these)
 - Ghorbani et al. 2019, 'Interpretation of Neural Networks is Fragile' - tiny perturbations change saliency while keeping the prediction. arXiv:1710.10547
 - Kindermans et al. 2019, 'The (Un)reliability of saliency methods'. arXiv:1711.00867
 - Dombrowski et al. 2019, 'Explanations can be manipulated and geometry is to blame'. arXiv:1906.07983
- Detecting adversarial examples
 - Feinman et al. 2017, 'Detecting Adversarial Samples from Artifacts'. arXiv:1703.00410
 - Xu et al. 2018, 'Feature Squeezing' - prediction disagreement under input transforms. arXiv:1704.01155
 - Carlini & Wagner 2017, 'Adversarial Examples Are Not Easily Detected' - caveat: adaptive attacks bypass many detectors. arXiv:1705.07263

Approach & implementation

- Model: torchvision ResNet-50 (ImageNet); all attacks/explanations run in $[0,1]$ pixel space with normalization folded into the model.
- Attacks (implemented from scratch, PyTorch): untargeted FGSM and L-inf PGD, pushing the model away from its own clean prediction.
- Explanations: Grad-CAM from scratch (forward/backward hooks on layer4) + Integrated Gradients via Captum (the library used by the course's reference ViT project).
- Explanation shift: cosine, Spearman rank, and top-10% region IoU between the clean and adversarial saliency maps for the same target class.
- Detector: 'prediction instability' = fraction of tiny Gaussian-noised copies whose label changes; scored by ROC-AUC. Forward-only, no training.
- Tooling: PyTorch, torchvision, Captum, scikit-learn, matplotlib. Run on Apple-Silicon MPS in ~ 40 s. Full code + figures in the repo (last slide).

EXPERIMENT A

Result A - attacks succeed with invisible noise



PGD reaches 100% success; FGSM is far weaker. The budget is imperceptible ($L_\infty \leq$ a few /255).

EXPERIMENT B

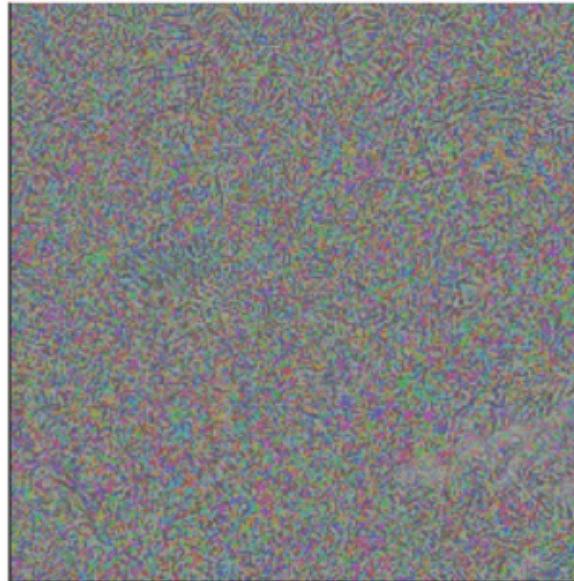
Result B - same picture, different label

Same picture to a human - different label to the model

prediction: giant panda
confidence 0.56



adversarial perturbation (x10)
 $L_{\infty} = 0.0157$



prediction: tennis ball
confidence 0.01

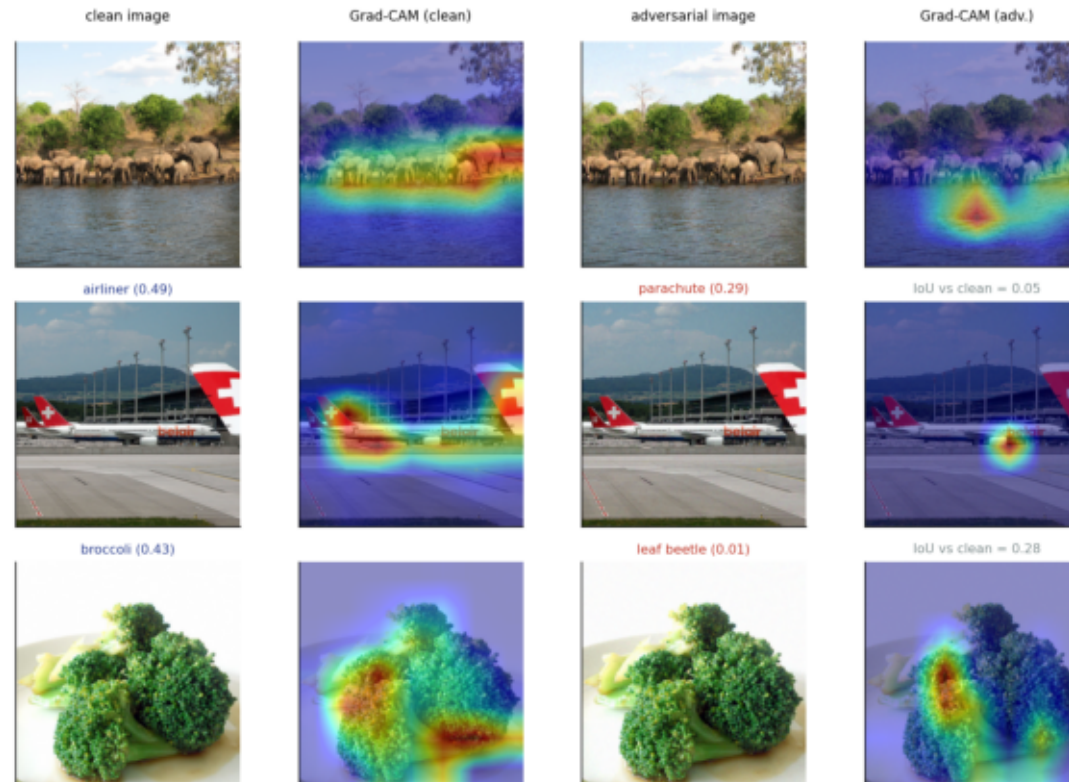


'giant panda' (0.56) -> 'tennis ball' (0.01) at $L_{\infty}=0.016$; the perturbation (centre, x10) is invisible at scale 1.

EXPERIMENT B

Result B - explanations relocate under attack

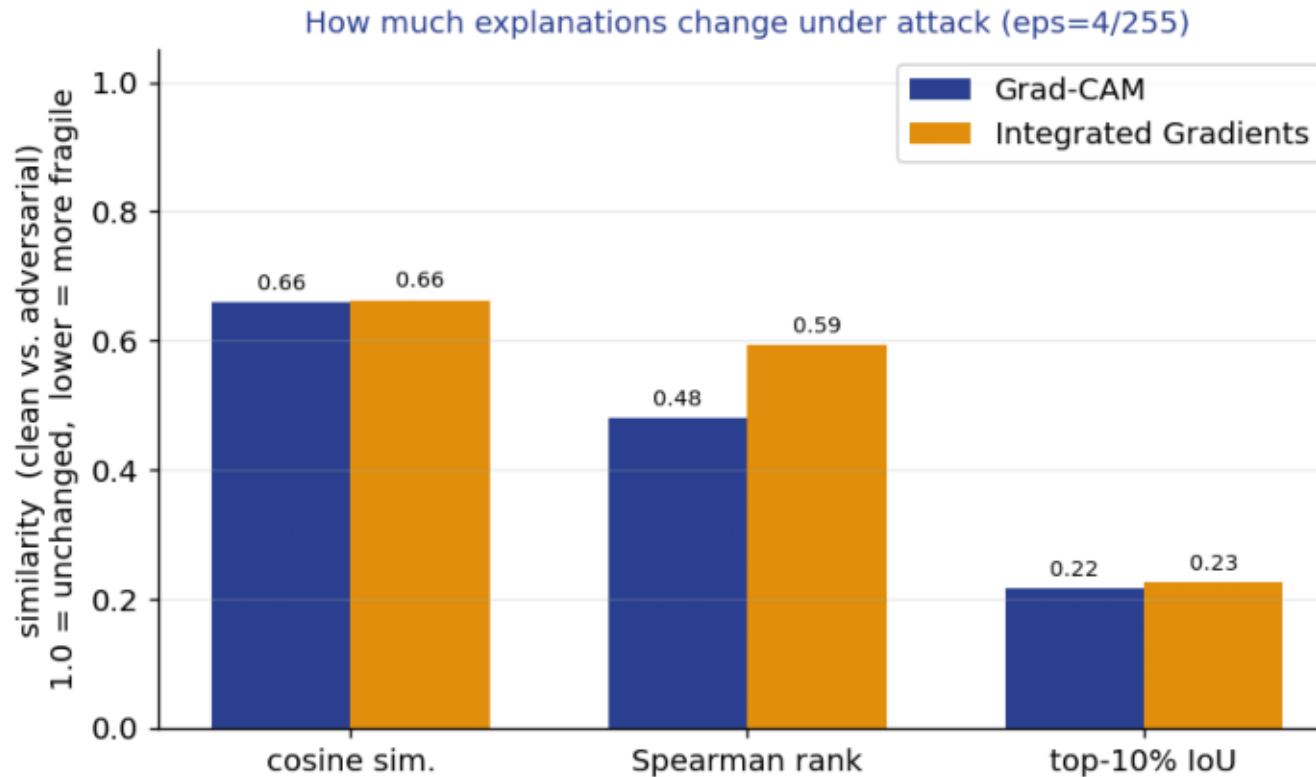
Explanations are fragile: an imperceptible attack relocates the saliency map



Grad-CAM moves off the object after an imperceptible perturbation - the 'explanation' no longer reflects the (now wrong) decision.

EXPERIMENT B

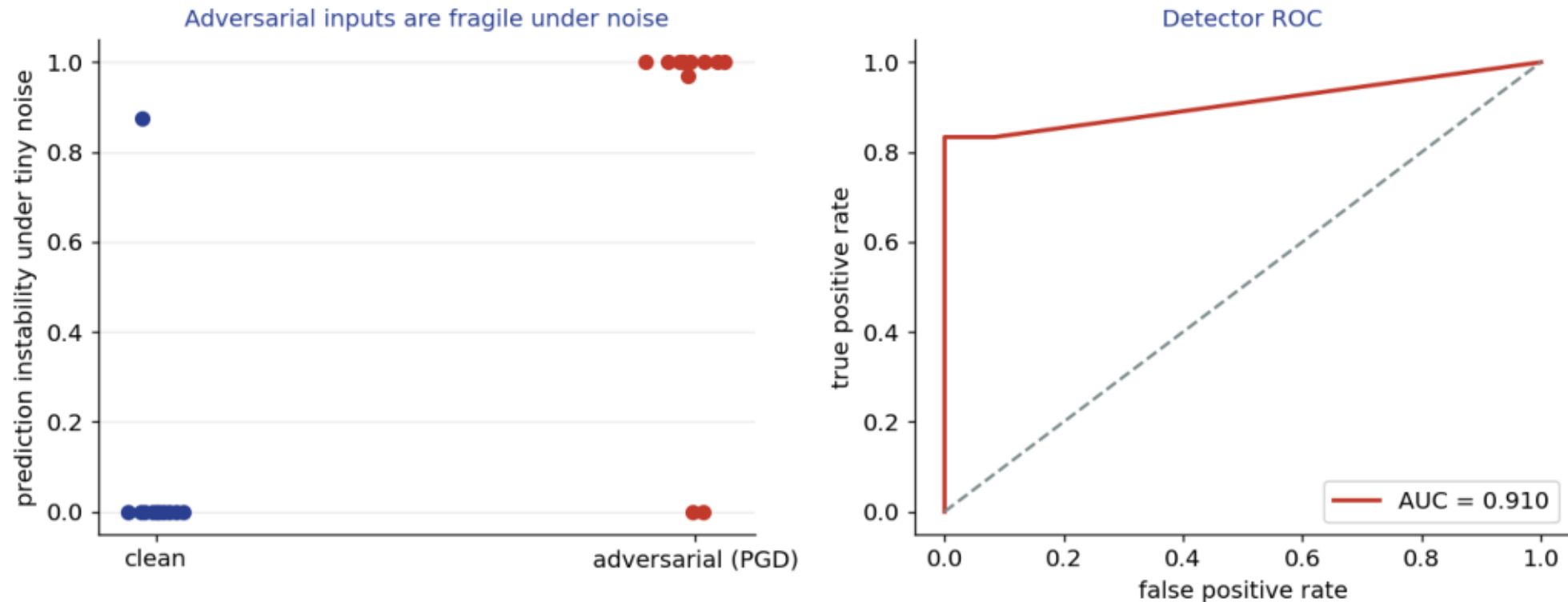
Result B - how fragile, quantitatively



Only ~22% of the top-10% salient region survives (Grad-CAM IoU 0.22, IG 0.23). Both methods are fragile.

EXPERIMENT C

Result C - fragility becomes a detector



Clean instability 0.07 vs adversarial 0.83 -> ROC-AUC = 0.91. The very fragility that breaks explanations flags attacks.

Discussion & limitations

- Explanations alone are not a security guarantee: an imperceptible attack flips the label AND the saliency map, so a user reading the heatmap would be misled.
- But behaviour under noise is informative: adversarial inputs sit near a decision boundary and are unstable, which our simple detector exploits (AUC 0.91).
- Limitations: small sample (12 images, single architecture); white-box, L-inf attacks only; the detector is non-adaptive.
- Carlini & Wagner (2017) caution: an adversary aware of the detector can craft noise-robust attacks - so this is a hurdle, not a solution.

Conclusion & future work

- Attacks succeed, explanations break, and the break is detectable - one pipeline links robustness (L07) and interpretability (L09).
- Headline numbers: PGD up to 100% success; explanation top-IoU ~ 0.22 ; detector AUC 0.91.
- Future work:
 - evaluate on a full validation set + more architectures (ViT vs CNN);
 - adaptive attacks that preserve the explanation / evade the detector;
 - adversarial training - do robust models also have robust explanations?

Links & references

- Code (all experiments, one command): <https://github.com/jagrat97/adversarial-attacks-vs-explanations>
- Demo video: https://youtu.be/REPLACE_WITH_YOUR_VIDEO_ID
- Reproduce: `python src/run_experiments.py` (figures + results.json)
- Live demo: `python src/demo.py --image zebra --attack pgd --eps 8/255`
- Key references: Goodfellow 2015 (FGSM, 1412.6572); Madry 2018 (PGD, 1706.06083); Selvaraju 2017 (Grad-CAM, 1610.02391); Sundararajan 2017 (IG, 1703.01365); Ghorbani 2019 (fragile interpretations, 1710.10547); Xu 2018 (feature squeezing, 1704.01155).
- Course lectures: github.com/fatheral/ai-intro-course (L07 Robustness, L09 Interpretability).

Thank you - questions?